

Kerr–de Sitter Cosmological Conjecture + CMB

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July 9... Added CMB Dataset to python Kerr-de Sitter Chebyshev Modelling.

1. Executive Summary

This document extends the current multi-segment Chebyshev modelling within a Kerr-de Sitter structure that has subsequently been extended to include Cosmic Microwave Background CMB data.

Cosmological structure to include outlines an expanded cosmological model based on Kerr-de Sitter space-time and rotating spacetime fabric (STF).

It offers a coherent structure that addresses known issues such as the Hubble tension, CMB anomalies, and the role of dark energy.

A Chebyshev polynomial-based expansion models the luminosity distance fit, and the rotational nature of space-time provides a compelling alternative explanation to the dark energy paradigm.

2. Planck CMB Spectrum Archive.

The Planck best-fit CMB temperature power spectrum, shown in the figure below, covers the wide range of multipoles $\ell = 2-2508$.

Over the multipole range $\ell = 2-29$, the power spectrum is derived from the "Commander" component-separation algorithm applied to the combination of Planck 2018 temperature data between 30 and 857 GHz, including 86% of the sky.

2.0 HTML indexes

https://wiki.cosmos.esa.int/planck-legacy-archive/index.php/CMB_spectrum_%26_Likelihood_Code#Best-fit_model

A CMB dataset provided by European Space Agency

<https://esdcdoi.esac.esa.int/doi/html/data/astronomy/planck/Cosmology.html>

2.1 Planck CMB Temperature Power Spectrum

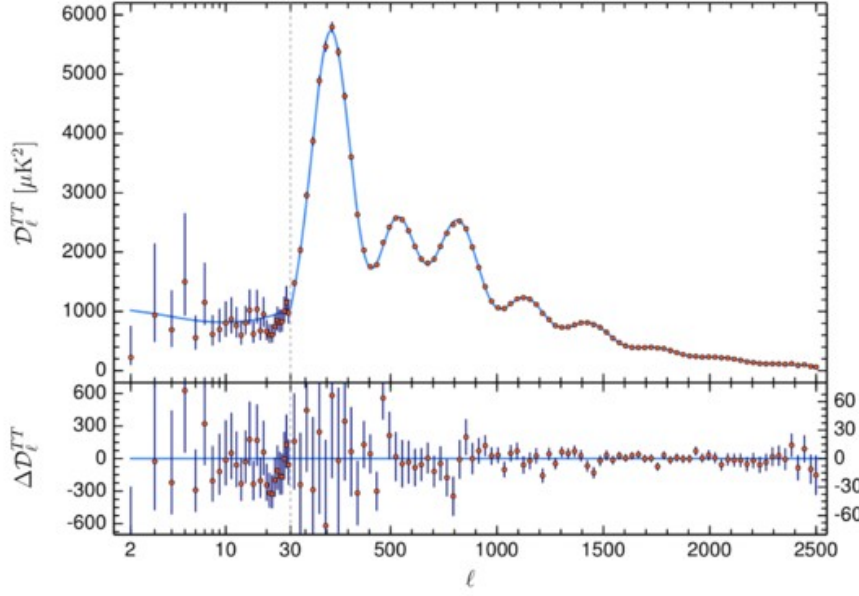
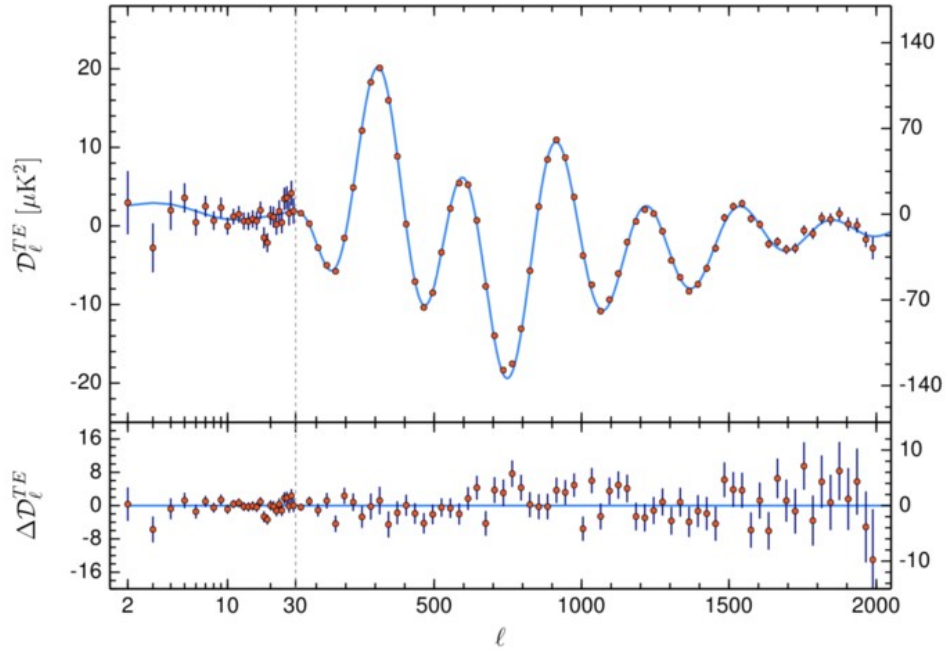


Fig.1. *Planck* 2018 temperature power spectrum. At multipoles $\ell \geq 30$ we show the frequency-coadded temperature spectrum computed from the *Planck* cross-half-mission likelihood, with foreground and other nuisance parameters fixed to a best fit assuming the base- Λ CDM cosmology. In the multipole range $2 \leq \ell \leq 29$, we plot the power spectrum estimates from the *Commander* component-separation algorithm, computed over 86 % of the sky. The base- Λ CDM theoretical spectrum best-fit to the *Planck* TT,TE,EE+lowE+lensing likelihoods is plotted in light blue in the upper panel. Residuals with respect to this model are shown in the lower panel. The error bars show $\pm 1\sigma$ diagonal uncertainties, including cosmic variance (approximated as Gaussian) and not including uncertainties in the foreground model at $\ell \geq 30$.

2.2 Planck 2012 CMB TE Power Spectra



2.3 Planck 2012 CMB EE Power Spectra

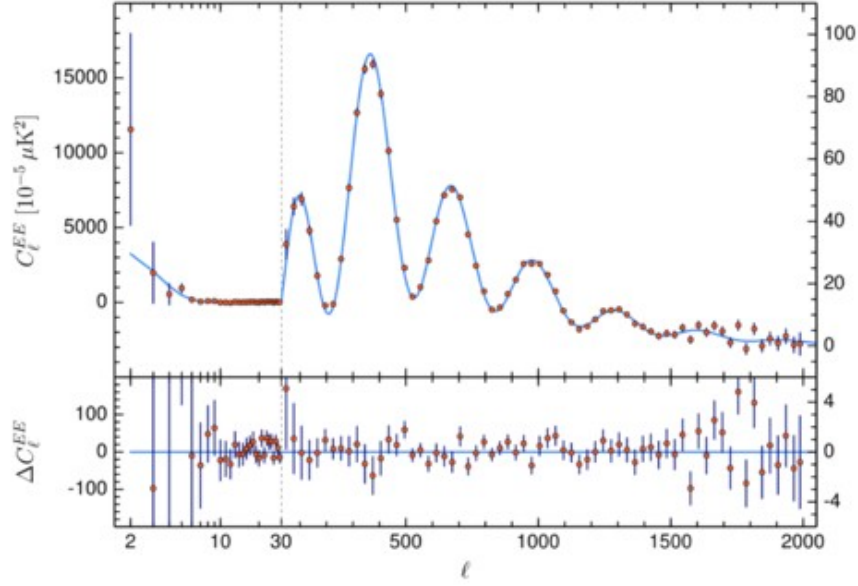


Fig. 2. *Planck* 2018 *TE* (top) and *EE* (bottom) power spectra. At multipoles $\ell \geq 30$ we show the coadded frequency spectra computed from the *Planck* cross-half-mission likelihood with foreground and other nuisance parameters fixed to a best fit assuming the base- Λ CDM cosmology. In the multipole range $2 \leq \ell \leq 29$, we plot the power spectra estimates from the *SimAll* likelihood (though only the *EE* spectrum is used in the baseline parameter analysis at $\ell \leq 29$). The best-fit base- Λ CDM theoretical spectrum fit to the *Planck* TT,TE,EE+lowE+lensing likelihood is plotted in light blue in the upper panels. Residuals with respect to this model are shown in the lower panels. The error bars show Gaussian $\pm 1\sigma$ diagonal uncertainties including cosmic variance. Note that the vertical scale changes at $\ell = 30$, where the horizontal axis switches from logarithmic to linear.

